TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 17

5-MINUTE REVISION

Uniformity Trials

Studying soil heterogeneity: fertility maps, serial correlation, strip mean squares, Smith's law and optimum plot size.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

Use uniformity trials to study field heterogeneity.

REMEMBER

Grow a uniform crop and measure small basic units before deciding useful plot sizes, shapes and blocks.

APPLY IT

If fertility changes down the field, orient blocks to contain relatively similar plots rather than mixing the full gradient within each block.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 What is a uniformity trial?

A trial to study the **nature and extent of fertility variation** in a field.

- One crop variety over the whole field; all inputs applied **uniformly**.
- Remove a border, then harvest separately by **basic units** (smallest recorded unit).
- Yield differences among basic units measure **soil heterogeneity**.

Helps decide: plot size, plot shape, block size and blocking direction.

2 Fertility contour map

A map of yields over the field. **Moving averages** smooth random variation; areas of similar fertility are joined by contour lines.

Rice example: 12 m × 17 m field, 1 m border removed → 10 m × 15 m → 150 basic units of 1 m × 1 m.

3 Serial correlation

Neighbouring units

$$r_s = \frac{\sum X_i X_{i+1} - \frac{(\sum X_i)^2}{n}}{\sum X_i^2 - \frac{(\sum X_i)^2}{n}}$$

Computed horizontally and vertically. High r_s: neighbouring yields are similar. Low r_s: little linear association at that lag (not proof of patches).

4 Mean square between strips

Vertical strips

$$MS_V = \frac{\frac{\sum V_i^2}{r} - \frac{G^2}{n}}{c-1}$$

Horizontal strips

$$MS_H = \frac{\frac{\sum H_i^2}{c} - \frac{G^2}{n}}{r-1}$$

A simpler method with the same aim as serial correlation. The larger mean square shows the direction of greater variation. $n = r \times c$.

5 Fairfield Smith's variance law

Variance law

$$V_x = \frac{V_1}{x^b}$$

Log form (linear)

$$\log V_x = \log V_1 - b \log x$$

Regression through origin

$$Y = cX, \quad c = -b$$

V_x = variance of yield per basic unit for plots of x units; b = **Smith's index of soil heterogeneity**, idealised range 0 to 1. $Y = \log V_x - \log V_1$, $X = \log x$. Example: $b = 0.3607$, adjusted ≈ 0.4116.

6 Simulated plots

Variance per unit area

$$V_x = \frac{Q_x}{rc - x}$$

Combine basic units into plots of x units ($w = rc/x$ plots, fitting the field exactly). A plot's yield is the **sum** of its units.

7 Maximum curvature method

Plot **plot size** (X-axis) against **coefficient of variation** (Y-axis). CV generally falls as plot size rises; the point of **maximum curvature** (the bend) is taken as the optimum plot size.

Larger plots	Caution
Lower CV, but fewer plots fit a fixed area.	The bend depends on axis scales and the available arrangements: treat it as a guide.

! Common mistakes

- ✗ Different shapes partition the same field, so their variances are **dependent**: no ordinary F test.
- ✗ High b alone doesn't prove a patch pattern.

? Fill in the blanks

1. If Smith's index b = 0, a larger plot ____ the variance per unit area. **[does not reduce]**
2. If b = 1, doubling the plot size ____ the variance per unit area. **[halves]**
3. The slope of log V_x plotted against log x equals ____. **[−b]**
4. A 10 m × 15 m net area cut into 1 m² units gives ____ basic units. **[150]**
5. In the maximum curvature method, CV is plotted against ____. **[plot size]**
6. In a uniformity trial every basic unit gets the ____ variety and management. **[same]**